

$$n = 100$$



Module 4 Chapter 7 Confidence Intervals

What is a confidence interval?

The purpose of collecting data on a sample is not simply to have data on that sample. Researchers take the sample in order to infer from that data some conclusion about the wider population represented by the sample.

The **confidence interval** or **CI** is the **range** in which a parameter (e.g. the mean value) lies with a certain **probability**. If several samples are taken from a population, it is very likely that each sample will have a different mean value. However, we want to know the **mean of the population**, not the mean of the sample. **The confidence interval is the range in which the true mean of the population lies with a certain probability.**

Word of Caution:

The above definition is widely used because it is easy to understand, but not all experts agree that it is correct. The following definition is definitely correct, but more complicated:

The 95% confidence interval (CI) is an interval calculated using sample data from an infinite series, 95% of which contain the population parameter. In the long run, 95% of such intervals contain the true mean.

To obtain confidence interval, we **add** and **subtract** the **margin of error** from the **sample mean**. This result gives us the **upper limit** and the **lower limit** of the **confidence interval**. The confidence interval may be wider or narrower depending on the degree of certainty, or estimation precision, that is required.

Confidence interval = sample mean $\pm$ margin of error

How to calculate margin of error?

$$\bar{x} \pm \text{margin of error}$$

$$CI = \left\{ \bar{X} \pm z \cdot \frac{\sigma}{\sqrt{n}} \right\}$$

- The margin of error depends on the ***degree of confidence*** that is required for the estimation.
- Typically, degrees of confidence vary between 90% and 99.9% (with 95% being most common), but it is up to the researcher to decide.
- The level of ***confidence*** is represented by ***z-value*** which can be found in the ***confidence table***.
- It is also necessary to know the ***standard deviation*** of the variable in the ***population***. (Note: the ***population standard deviation*** is ***NOT*** the same as the ***sample standard deviation***).
- The ***size of the sample n*** will be used to compute the margin of error.

$$\text{Margin of Error} = z \times \frac{\text{population standard deviation}}{\sqrt{n}} = z \times \frac{\sigma}{\sqrt{n}}$$

Let's take an example and understand better

Suppose we're measuring the average level of conscientiousness in the U.S. population, a psychological trait that varies and can be measured with a personality test. If we measured just one person with a test, we should have very little confidence that the "average" of that value is an accurate mean for the U.S.; our sample size was terrible. But as we average more individuals – ten, thirty, a thousand – we might start to feel that our average was a more accurate representation of the U.S. population.

Simple conditions for inference about the mean

There are certain requirements that must be met before making inferences about a population mean:

- The sample must be ***randomly*** selected.
- The variable of interest must have a ***Normal distribution*** $N(\mu, \sigma)$ in the population.

- The population mean μ is **unknown**, but the standard deviation σ for the variable must be **known**.

These conditions are very difficult to meet in a real situation, especially in social science research. There are other procedures that need to be followed when these conditions are not met.

Let us first start with the best-case scenario, when all the above conditions are met!

Here's the data collected for the mean conscientiousness score:

Sample size = 400

Sample mean = 80

Population Standard Deviation = 40

Confidence Level = 95%

We know, to construct the confidence interval, we use:

Confidence interval = sample mean $\pm$ margin of error

Let's calculate margin of error:

$$\text{Margin of Error} = z \times \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{40}{\sqrt{400}} = 3.92$$

So, our confidence interval becomes:

$$\{\bar{x} - 3.92, \bar{x} + 3.92\} = \{76.08, 83.92\}$$

A t-interval for the mean

So far, we have assumed the following:

- The population standard deviation σ is **known**, and
- $X_1, X_2, \dots, X_n$ are **normally distributed**. (The truth is that $X_1, X_2, \dots, X_n$ **need not** be normally distributed as long as the sample size n is **large enough** for the Central Limit Theorem to apply. In this case, the confidence interval is an **approximate** confidence interval.)

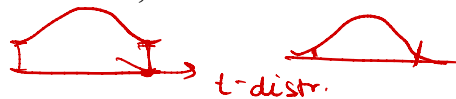
It is unrealistic to think that we'd ever be in a situation where the first condition would be met. That is, when would we ever know the population standard deviation σ , but not the population mean μ ? Let's entertain, then, the realistic situation in which not only the population mean μ is unknown, but also the population standard deviation σ is **unknown**.

The reasonable thing to do is to **estimate** the population standard deviation σ with the sample standard deviation (using a process called **Bessel's correction**): (Since we are **estimating** the population standard deviation using sample standard deviation, we no longer use σ . **The unbiased estimate of the population standard deviation is represented using S .**)

$$S = \sqrt{\sum_{i=1}^n \frac{1}{(n-1)} (X_i - \bar{X})^2}$$

$\sqrt{\sum_{i=1}^n \frac{(X_i - \bar{X})^2}{(n-1)}}$

Here, $(n-1)$ is called the “**Degrees of freedom**” and the distribution is a **t-distribution**. As the degrees of freedom increase, the t-distribution starts looking like a normal distribution.



Hence, our formula for margin of error now becomes:

$$\text{Margin of Error} = t \times \frac{\text{unbiased estimate of } \sigma}{\sqrt{n}} = t \times \frac{S}{\sqrt{n}}$$

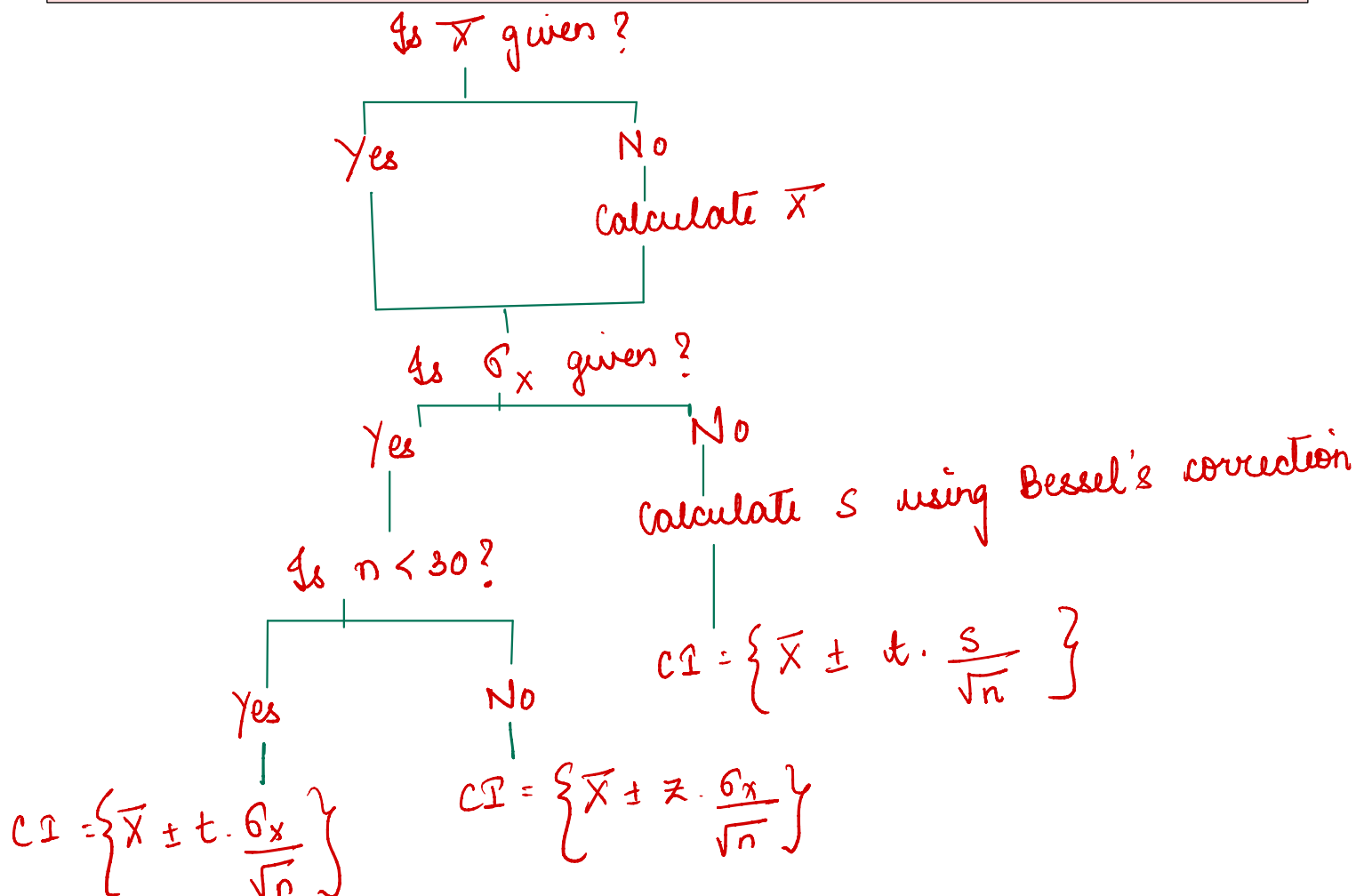
We can get the t -value for a given level of **confidence** and **degrees of freedom**, using the **t -table**.

We then use this margin of error to calculate our Confidence interval.

Let's Practice

- You want to rent an unfurnished studio apartment in NYC next year. The mean monthly rent for a random sample of 60 apartments advertised on Craig's list is \$1000. Assume a standard deviation of \$200. Construct a 95% confidence interval.
- How large of a sample of studio apartments would you need to estimate the population mean within ± 50 with 90% confidence?

- The monthly sales of an appliance shop are distributed normally with a standard deviation of \$900. A study of the sales in the last 9 months has found the confidence interval of the mean monthly sales with extremes of \$4663 and \$5839. What is the confidence of this interval?
- The quantity of hemoglobin in the blood stream of a man follows a normal distribution with standard deviation of 2g/dl. Calculate the confidence ~~interval~~ ^{level} for a sample of 12 men which indicates that the population mean blood hemoglobin is between 13 and 15g/dl.
- The mean emission of all engines of a new design needs to be below 20 ppm if the design is to meet new emission requirements. 10 engines are manufactured for testing purposes and the emission level of each is determined. The emission data is as follows:
15.6, 16.2, 22.5, 20.5, 16.4, 19.4, 16.6, 17.9, 12.7, 13.9
Does the data supplied above provide evidence that the new engines meet the standard?



1.

Here, for 95%, our z value is ± 1.96

$\therefore \mu$ lies in the range of

$$\left(\bar{X} \pm z \cdot \frac{\sigma_x}{\sqrt{n}} \right)$$

$$\mu = \bar{X} \pm \text{Margin of error}$$

$$\text{Margin of error} = z \cdot \frac{\sigma_x}{\sqrt{n}}$$

$\therefore \mu$ lies in the range $\rightarrow$

$$\left(\$1000 - 1.96 \times \frac{200}{\sqrt{60}}, \$1000 + 1.96 \times \frac{200}{\sqrt{60}} \right)$$

$$\approx (\$949.39, \$1050.61)$$

2.

Margin of error = 50, confidence 90%.

$$\therefore 50 = 1.645 \times \frac{\sigma_x}{\sqrt{n}} = \frac{1.645 \times 200}{\sqrt{n}}$$

$$\Rightarrow n = \left(\frac{1.645 \times 200}{50} \right)^2 = (6.580)^2 \approx 43.3$$

Round off to
the next whole
number

$$\therefore n = 44$$

3.



we need to calculate our z values and then look at the corresponding confidence interval

$$\bar{X} + z \cdot \sigma_{\bar{X}} = 5839 \quad \text{(i)}$$

$$\bar{X} - z \cdot \sigma_{\bar{X}} = 4663 \quad \text{(ii)}$$

subtracting eq(ii) from eq. (i)

$$2 \cdot z \cdot \sigma_{\bar{X}} = 5839 - 4663$$

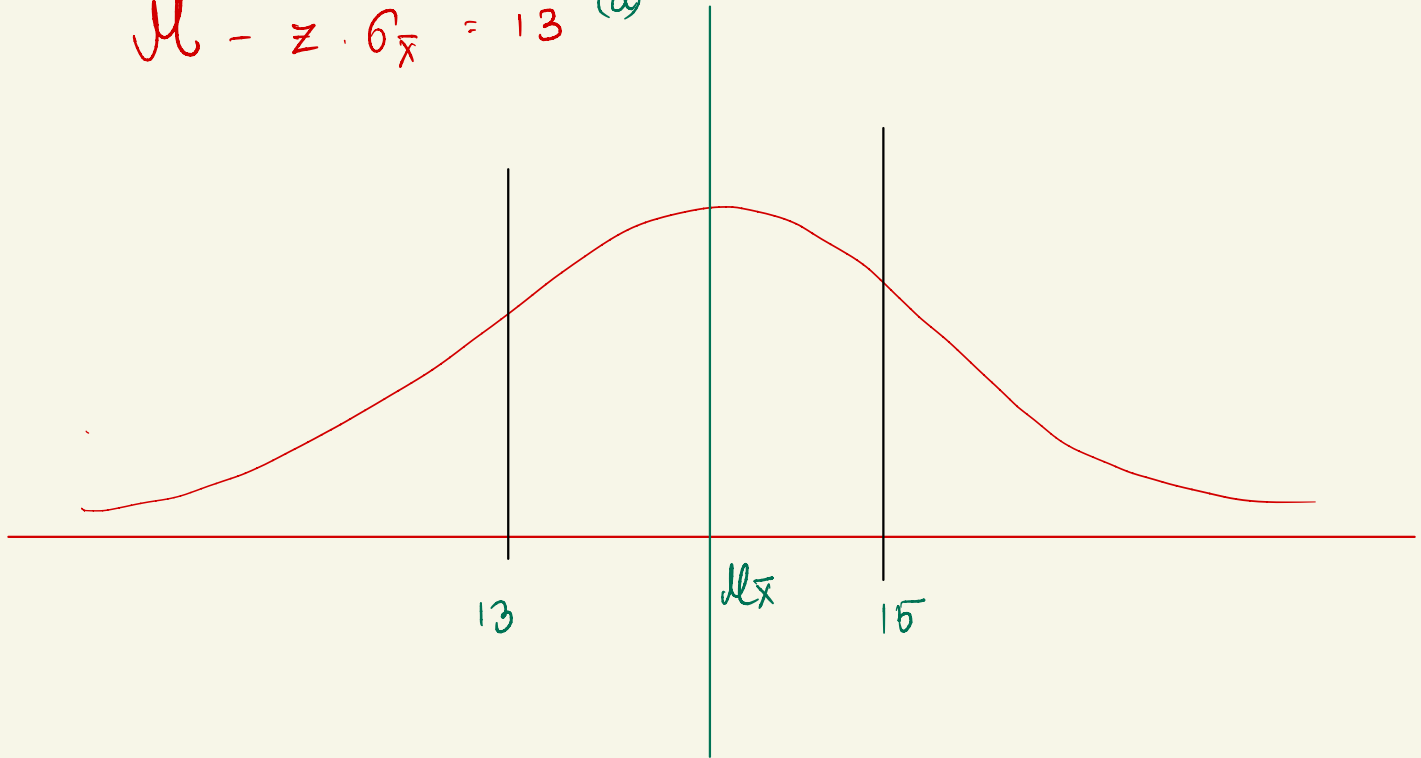
$$\Rightarrow z = \left(\frac{5839 - 4663}{2} \right) \times \frac{\sqrt{n}}{\sigma_x}$$

$$= \frac{588 \times \sqrt{9}}{900} = \frac{588 \times 3}{900} \approx 1.96$$

for $z=1.96$, confidence is 95%

$$4. \quad \mu + z \cdot \sigma_{\bar{x}} = 15 \quad (i)$$

$$\mu - z \cdot \sigma_{\bar{x}} = 13 \quad (ii)$$



Subtracting eq (ii) from eq (i)

$$2z \cdot \sigma_{\bar{x}} = 15 - 13$$

$$z = \frac{15 - 13}{2 \sigma_{\bar{x}}} = \frac{2}{2 \sigma_{\bar{x}}} = \frac{\sqrt{n}}{\sigma_x} = \frac{\sqrt{12}}{2} \approx 1.73$$

$\Rightarrow$ 92% confidence interval

(real value is 91.64%)

5. $n = 10 < 30$ $\therefore$ t -distribution

σ_x is not known. $\therefore$ we use Bessel's correction to calculate s using sample data

we calculate $\bar{X} = 17.17$

$$s = \sqrt{\frac{1}{9} \left\{ (15.6 - 17.17)^2 + (16.2 - 17.17)^2 + \dots + (13.9 - 17.17)^2 \right\}}$$

$$= 2.98$$

degrees of freedom $\rightarrow 9$, Confidence $\rightarrow 95\%$

$$\therefore t\text{-value} = 2.262$$

To check if the CI is below 20 or not

$$CI = \left\{ 17.17 \pm 2.262 \times \frac{2.98}{\sqrt{10}} \right\}$$

$$= \{ 15.04, 19.3 \}$$

there is a 95% probability $\mu < 20$.

$\therefore$ Yes, engine meets standard!

How much greater will t -value be w.r.t.
corresponding z -value?

This depends on the degrees of freedom: $(n-1)$ where
 n is the sample size.

n	df	t
6	5	2.571
11	10	2.228
31	30	2.042
51	50	2.009
101	100	1.984
∞	∞	1.960

← t values at 95% CP

→ Same as z value at 95% CP

Sometimes, it's generalized and assumed that if
 $n \geq 30$, one can directly use z -value.

(Although Questionable!!!)

Confidence Level	z
0.70	1.04
0.75	1.15
0.80	1.28
0.85	1.44
0.90	1.645
0.92	1.75
0.95	1.96
0.96	2.05
0.98	2.33
0.99	2.58

Two-tail probability
One-tail probability

0.20 0.10 0.05 0.02 0.01
0.10 0.05 0.025 0.01 0.005

Confidence levels

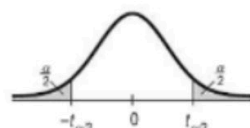
80% 90% 95% 98% 99%

Table T

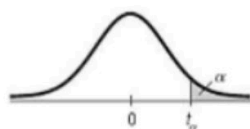
df

Values of t_{α}

df



Two tails



One tail

1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5
6	1.440	1.943	2.447	3.143	3.707	6
7	1.415	1.895	2.365	2.998	3.499	7
8	1.397	1.860	2.306	2.896	3.355	8
9	1.383	1.833	2.262	2.821	3.250	9
10	1.372	1.812	2.228	2.764	3.169	10
11	1.363	1.796	2.201	2.718	3.106	11
12	1.356	1.782	2.179	2.681	3.055	12
13	1.350	1.771	2.160	2.650	3.012	13
14	1.345	1.761	2.145	2.624	2.977	14
15	1.341	1.753	2.131	2.602	2.947	15
16	1.337	1.746	2.120	2.583	2.921	16
17	1.333	1.740	2.110	2.567	2.898	17
18	1.330	1.734	2.101	2.552	2.878	18
19	1.328	1.729	2.093	2.539	2.861	19
20	1.325	1.725	2.086	2.528	2.845	20
21	1.323	1.721	2.080	2.518	2.831	21
22	1.321	1.717	2.074	2.508	2.819	22
23	1.319	1.714	2.069	2.500	2.807	23
24	1.318	1.711	2.064	2.492	2.797	24
25	1.316	1.708	2.060	2.485	2.787	25
26	1.315	1.706	2.056	2.479	2.779	26
27	1.314	1.703	2.052	2.473	2.771	27
28	1.313	1.701	2.048	2.467	2.763	28
29	1.311	1.699	2.045	2.462	2.756	29
30	1.310	1.697	2.042	2.457	2.750	30
32	1.309	1.694	2.037	2.449	2.738	32
35	1.306	1.690	2.030	2.438	2.725	35
40	1.303	1.684	2.021	2.423	2.704	40
45	1.301	1.679	2.014	2.412	2.690	45
50	1.299	1.676	2.009	2.403	2.678	50
60	1.296	1.671	2.000	2.390	2.660	60
75	1.293	1.665	1.992	2.377	2.643	75
100	1.290	1.660	1.984	2.364	2.626	100
120	1.289	1.658	1.980	2.358	2.617	120
140	1.288	1.656	1.977	2.353	2.611	140
180	1.286	1.653	1.973	2.347	2.603	180
250	1.285	1.651	1.969	2.341	2.596	250
400	1.284	1.649	1.966	2.336	2.588	400
1000	1.282	1.646	1.962	2.330	2.581	1000
∞	1.282	1.645	1.960	2.326	2.576	∞

Confidence levels

80% 90% 95% 98% 99%

Two-tail probability

0.20 0.10 0.05 0.02 0.01

One-tail probability

0.10 0.05 0.025 0.01 0.005